

LAST NAME: \_\_\_\_\_

## Practice Quiz 10A — MATH 2055 — Fall 2026

### Practice for Quiz 10

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

*Auto-generated by the Codex harness using GPT-5.*

1. (a) Let  $A = \{1, 2, 3, 4\}$ ,  $B = \{a, b, c\}$ , and define

$$f(1) = a, \quad f(2) = b, \quad f(3) = b, \quad f(4) = c.$$

For  $X = \{1, 3\}$ , compute  $f(X)$  and  $f^{-1}(f(X))$ . Are the latter and  $X$  equal?

- (b) State the geometric sum formula for  $1 + r + \cdots + r^n$ , including its hypotheses, and prove it by induction on  $n$ .